THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 2: **Arithmetic Expressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. 8 × 91 + 8 × 9 is equal to 8 × ?. The missing number is: 1

(a) 108 **(b)** 99 **(c)** 101 **(d)** 100

2. A cyclist covers 40 km in the first hour and 5 km less every following hour for 6 hours in all. The total distance (km) is: 1

(a) 164 **(b)** 170 **(c)** 166 **(d)** 165

3. The value of $(72 \div (9 - 3) + 4 \times (5 + 2)) \div 10$ is: 1

(a) 4 **(b)** 6 **(c)** 2 **(d)** 14

4. The value of $(64 - 4 \times (9 + 3)) \div (2 \times 4)$ is: 1

(a) 2 **(b)** 1086 **(c)** 34 **(d)** 4

5. The value of $(9 \times 8 - 4 \times (6 + 3)) \div (18 \div 3)$ is: 1

(a) 8 **(b)** 6 **(c)** 16 **(d)** 4

6. The value of 'twice the sum of 9 and 11, decreased by 13' is: 1

(a) 14 **(b)** 27 **(c)** 40 **(d)** 53

7. Compare 25 × 16 and 24 × 17. The first minus the second is: 1

(a) -8 **(b)** -6 **(c)** -9 **(d)** -7

8. The value of 78 × 99 can be found as 78 × 100 - 78 = : 1

(a) 7722 **(b)** 7800 **(c)** 7822 **(d)** 7644

9. **Assertion (A):** $8 \times 99 = 792$. 1

Reason (R): $99 = 100 - 1$, so $8 \times 99 = 8 \times 100 - 8 \times 1$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** $6 + 2 \times (9 - 4) = 16$. 1

Reason (R): The brackets are worked out first.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Without finding the products, compare 30 × 17 and 29 × 18. Check by computing. 2

12. Evaluate $(9 \times 8 - 4 \times (6 + 3)) \div (18 \div 3)$. 2

13. Use the distributive property to find 18 × 45 + 18 × 55. 2

14. Without finding the products, compare 18 × 12 and 17 × 13. Check by computing. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A shop sells 48 notebooks at ₹32 each and 36 pens at ₹14 each, and gives a discount of ₹40 on every 8 notebooks. Write an expression for the total amount and find it (discount counted once for each full set of 8 notebooks). 3

16. Evaluate 11×17 and $(11 + 2) \times (17 - 2)$ and find the difference between the two values. 3

17. A rectangular garden is 36 m long and 20 m wide. A path of width 3 m runs around outside. Write an expression for the area of the path and evaluate it. 3

OR

A shop sells 42 notebooks at ₹38 each and 24 pens at ₹16 each, and gives a discount of ₹35 on every 7 notebooks. Write an expression for the total amount and find it (discount counted once for each full set of 7 notebooks). 3

18. Evaluate 13×19 and $(13 + 3) \times (19 - 3)$ and find the difference between the two values. 3

OR

A rectangular garden is 45 m long and 30 m wide. A path of width 5 m runs around outside. Write an expression for the area of the path and evaluate it. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A teacher gives the expression $8 + 4 \times 3 - 2$ and asks students to place brackets to change its value. 5

(i) Find the value without brackets.
(ii) Find the value of $(8 + 4) \times 3 - 2$.
(iii) Find the value of $8 + 4 \times (3 - 2)$.
(iv) Which of the three has the greatest value and how much more is it than the value without brackets?

20. A school fete: 55 children each buy 5 tickets at ₹6 each. 25 adults buy 4 tickets at ₹28 each. The organisers spend ₹600 on prizes and ₹350 on decoration. Write one expression for their profit and evaluate it. 5

OR

Evaluate $(40 - 8) \times 2$ and $40 - (8 \times 2)$. Which bracketing gives the larger value and by how much? 5

END OF QUESTION PAPER — Total: 40 Marks